Exercise 3.5

1. Shane has $\frac{4}{5}$ of a bar of chocolate. He shares it equally between two people. What fraction of the bar do they each get?

2. Find the reciprocal of each of the following fractions:

(i) $\frac{7}{3}$ (ii) $\frac{4}{9}$ (iii) 6 (iv) $\frac{1}{5}$ (v) $\frac{3}{11}$

3. Find the reciprocal of each of the following fractions:

(i) $2\frac{1}{3}$ (ii) $1\frac{2}{9}$ (iii) 10 (iv) $3\frac{4}{5}$ (v) $1\frac{1}{4}$

4. Rewrite each of these as a multiplication and then work out the answer.

(i) $\frac{3}{4} \div \frac{1}{2}$ (ii) $\frac{5}{6} \div \frac{2}{3}$ (iii) $\frac{2}{5} \div \frac{9}{10}$ (iv) $\frac{7}{12} \div \frac{1}{6}$

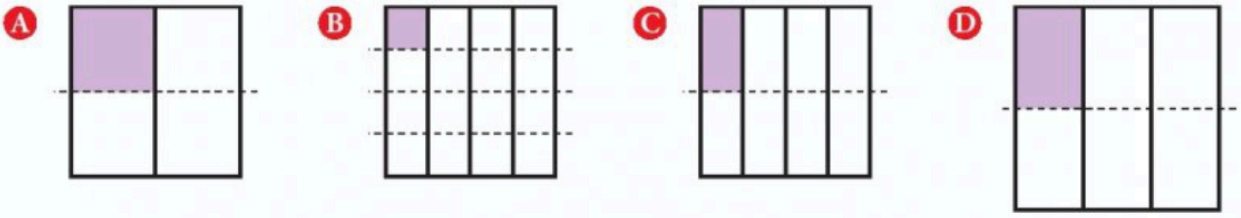
(v) $6 \div \frac{3}{4}$ (vi) $12 \div \frac{4}{9}$ (vii) $16 \div \frac{8}{9}$ (viii) $27 \div \frac{3}{4}$

5. Work out each of the following:

(i) $3\frac{3}{4} \div \frac{3}{8}$ (ii) $2\frac{5}{8} \div \frac{3}{4}$ (iii) $2\frac{1}{10} \div \frac{3}{5}$ (iv) $2\frac{5}{8} \div \frac{7}{16}$

(v) $1\frac{1}{8} \div 2\frac{1}{4}$ (vi) $8\frac{1}{4} \div 1\frac{3}{8}$ (vii) $5\frac{5}{8} \div 6\frac{1}{4}$ (viii) $3\frac{1}{8} \div 3\frac{3}{4}$

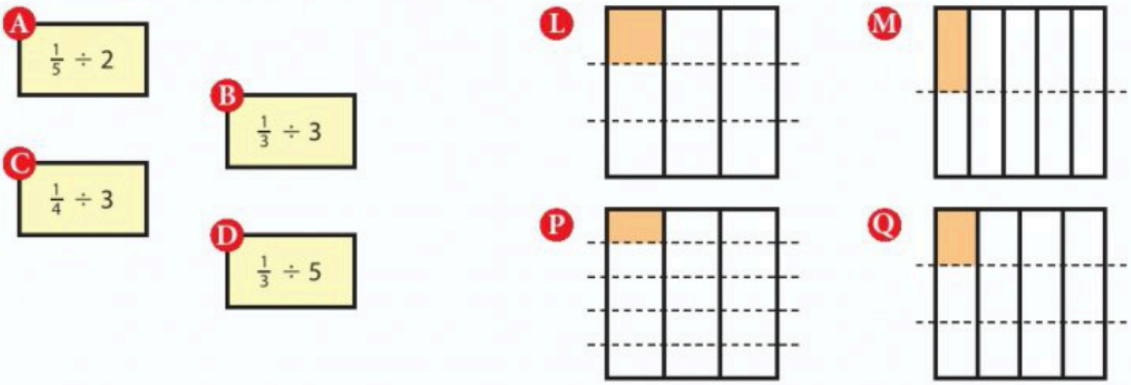
6. Which diagram below matches $\frac{1}{4} \div 2$?



7. Work out each of these:

(i) $5\frac{3}{7} \div 1\frac{3}{7}$ (ii) $10\frac{5}{6} \div 3\frac{1}{4}$ (iii) $6\frac{2}{3} \div 2\frac{4}{9}$ (iv) $1\frac{4}{5} \div \frac{27}{10}$

8. Match each calculation to a diagram.

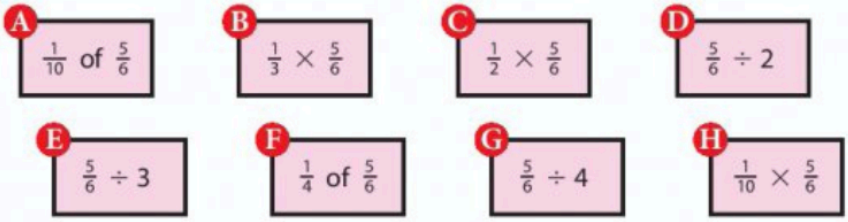


9. Simplify each of the following by first working out what is inside the brackets:

(i) $\left(\frac{3}{4} \times \frac{2}{9}\right) \div \frac{1}{12}$ (ii) $1\frac{1}{3} \times \left(\frac{3}{10} \div \frac{1}{2}\right)$ (iii) $\left(\frac{3}{4} \div \frac{7}{12}\right) \times 4\frac{2}{3}$

10. Denise has $\frac{1}{3}$ of a cake.
She shares it equally between herself, her brother and her sister.
What fraction of the cake do they each get?

11. Put these calculations into pairs which give the same answer.
Write down the answer to each pair.



12. Simplify each of the following by first working out what is inside the brackets:

(i) $\left(\frac{3}{5} + \frac{1}{3}\right) \div \frac{2}{3}$ (ii) $\left(\frac{4}{9} + \frac{1}{3}\right) \times \frac{6}{7}$ (iii) $\frac{5}{8} \div \left(3\frac{1}{2} - 1\frac{1}{4}\right)$

13.

S	G	R	I	O	A	D	P	W	L	E	M
$\frac{1}{12}$	$\frac{1}{10}$	$\frac{1}{9}$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{3}{16}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{3}{4}$

Work out each answer and use the code to change it to a letter.
Then rearrange each set of letters to spell a city.

(i) $\frac{4}{5} \div 2$ $\frac{1}{2}$ of $\frac{2}{9}$ $\frac{1}{2}$ of $1\frac{1}{2}$ $\frac{5}{6} \div 5$

(ii) $\frac{1}{10} \times \frac{5}{6}$ $\frac{1}{3}$ of $\frac{3}{4}$ $\frac{1}{16}$ of 3 $\frac{3}{8} \div 3$ $2 \times \frac{1}{18}$

(iii) $\frac{3}{5} \div 3$ $\frac{1}{2} \times \frac{2}{5}$ $\frac{1}{4}$ of 3 $\frac{1}{3} \times \frac{1}{3}$ $\frac{3}{8} \div 2$ $\frac{1}{4} \times \frac{1}{2}$

14. How many shelves could Carl make with $2\frac{1}{3}$ m of wood if each shelf measures $\frac{1}{6}$ m.

15. Show that if you divide $\frac{2}{3}$ m by $\frac{3}{4}$ m you get $\frac{8}{9}$. Interpret this result.